

TEMPLATE

KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

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VERSION **v. 1.2**

RELATED SUPPORT

- TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2

This document contains the following Sections

Key Project Information

SECTION A – Description of project

SECTION B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

SECTION C – Duration and crediting period

SECTION D – Summary of Safeguarding Principles and Gender Sensitive Assessment

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Appendix 1 – Safeguarding Principles Assessment (mandatory)

Appendix 2 - Contact information of Project participants (mandatory)

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KEY PROJECT INFORMATION

GS ID of Project	GS11209
Title of Project	Production and sale of efficient cookstoves in the urban areas of Maputo Province
Time of First Submission Date	23/06/2021
Date of Design Certification	27/05/2022
Version number of the PDD	07
Completion date of version	13/03/2023
Project Developer	Carbonsink Group S.r.l (Carbonsink)
Project Representative	Carbonsink Group S.r.l (Carbonsink)
Project Participants and any communities involved	N/A
Host Country (ies)	Mozambique
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☐ Small Scale ☒ Large Scale
Other Requirements applied	Community Services Activity Requirements (version 1.2)
Methodology (ies) applied and version number	Technologies and Practices to Displace Decentralized Thermal Energy Consumption Version 3.1
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☒ Regular ☐ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6.)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Emission Reductions	112,437	VERs

1 No Poverty	Average household savings i.e., decrease in expenditure on basic service such cooking, lighting, drinking	7,659	MZN/hh.y
7 Ensure access to affordable energy	Increased access to energy	26,397	households
15 Life On Land	Amount of non-renewable fuel savings	10,331	tons/year

SECTION A. DESCRIPTION OF PROJECT

A.1 Purpose and general description of project

The Project Activity “Production and sale of efficient cookstoves in the urban areas of Maputo Province” is implemented by Carbonsink.

In the urban areas, like in the cities of Maputo and Matola, of Maputo Province (Mozambique), charcoal is the main fuel used for cooking activities¹. Moreover, traditionally families use cookstoves with low energy efficiency which implies high consumption of charcoal. This means that a great deal of household incomes is spent for fuel purchasing and that there is an exposure to smoke during cooking activities especially to women and children who are responsible of cooking activities. Charcoal combustion and production is, moreover, a significant source of greenhouse gas (GHG) emissions responsible for climate change.

The project aims to address these issues by distributing fuel-efficient cookstoves in the urban areas of the Province of Maputo. The stove distribution is foreseen to be made mainly as door-to-door selling and it has started in August 2020 in the city of Maputo and the final number of the distributed under this project activity is estimated to reach

¹ Baumert et.al, 2016: Charcoal supply chains from Mabalane to Maputo: Who benefits? (Available at: <https://www.sciencedirect.com/science/article/pii/S0973082615302428>), Atanassov et al., 2012: Urban Biomass Energy Analysis Mozambique Ministry of Energy, Maputo (Available at: http://greenlight-africa.com/assets/final_report_mozambique_urban_biomass.pdf)

around 50,000² fuel-efficient cookstoves. The cookstove model distributed is locally produced charcoal fired model “Mbaula Poupa+”.

Besides reducing fuel consumption during the cooking activities and thus reducing greenhouse gas emission and contributing to climate change mitigation in line with the UN’s Sustainable Development Goals, the project is foreseen to provide several SDG benefits for the local families and communities.

A.1.1. Eligibility of the project under Gold Standard

The Gold Standard general eligibility criteria are documented in the Gold Standard for the Global Goals (GS4GG) Principle and requirements (version 1.2) specifically in section 3. According to Gold Standard for the Global Goals (GS4GG) Principle and requirements (version 1.2) para 4.1.3: “A Project type is automatically eligible for Gold Standard Certification if there are Gold Standard approved Activity Requirements and/or Impact Quantification Methodologies associated with it or it’s referenced in the Gold Standard Product Requirements. There is a methodology associated with the activity: “Technologies and Practices to Displace Decentralized Thermal Energy Consumption v3.1.” and the GS has published the Community Services Activity Requirements which include end-use energy efficiency projects, under which the project activity falls. Hence, the project activity falls under the automatic eligibility list of projects as the proposed project aims at distributing Improved Cook Stoves.

Eligibility Criterion	Means of Verification/Supporting evidence for inclusion
a) Type of project: Eligible Projects shall include physical action/implementation on the ground. Pre-identified eligible Project types are identified in the Eligibility Principles and Requirements section.	The projects involve the distribution of Improved Cookstoves (ICS). The project type is eligible under Community Services Activity Requirements .3.1.1(b)
b) Location of project: Projects may be located in any part of the world.	This project is located inside the boundaries of Mozambique. See section A.2 below.

² The here presented number of stoves to be included in this project activity is an estimation and it may vary based on the future distribution timelines.

c) Project area, boundary and scale: Section 3.1.2 (b): The Project Area and Project Boundary shall be defined. [...]and Products Requirements. In order to avoid double counting the [...]of impacts amongst projects).	The proposed project activity is a large scale energy efficiency project activity, and the project area as well as the boundaries are clearly defined in A.1. and section A.2 The project takes place inside geographical boundaries of Mozambique. The project is estimated to achieve an annual average of 112,437 tCO₂ emission reduction and thus qualify as a large scale project This project is not included by any other carbon standard neither voluntary or compliant. The avoiding of the double counting is confirmed in two ways: - The relevant online carbon registries have been consulted to confirm that the proposed activity is neither registered as a standalone project activity or as a part of a PoA with GS or other carbon standard. - Each project cookstove will be identified with unique ID number which enables the unique identification of the cookstove to be part of this project activity.
d) Host country requirements: Projects shall be in compliance with applicable Host Country's legal, environmental, ecological and social regulations.	This project is in compliance with Mozambique's legal, environmental, and social regulations. The project is actively contributing to the Host Country by reducing the GHG emissions, fuel consumption as well as economic and time savings to thousands of poor households.
e) Contact Details: As part of the Project Documentation the Project Developer shall provide (i) name and (ii) contact details [...]to refuse use of the Standard where reputational concerns are highlighted.	All the project participant information have been provided in Appendix 2.
f) Legal ownership Full and uncontested legal ownership[...] All projects shall immediately report to Gold Standard any land title/tenure disputes arising.	For each distribution, a Carbon rights transfer assertion will be signed and uploaded to our database stating that the rights to the carbon credits will lie with the project developer would persist. The elaborated arguments to demonstrate the additionality in line with this criterion are provided in the section B.5 of this PDD.
g) Other Rights: As well as legal title and ownership [...] Project implementation in affected areas.	There are no disputes or contested rights that have been identified in relation to rights relevant to the project activity.

h) Official Development Assistance (ODA) Declaration: All Project Developers applying for project activities [...] shall declare the Official Development Assistance (ODA) support.	It is confirmed that no Official Development Assistance (ODA) has been requested and none will be applied for the project activity.
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Eligibility criteria as per GS4GG Community Services Activities Requirements

Eligibility Criterion	Means of Verification/Supporting evidence for inclusion
Eligible project types	
2.1.2) All CSA projects shall lead to climate change mitigation and/or adaption by providing or improving access to services/resources at household or community level or institutional level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.	By providing improved cooking stoves to communities, the project improves access to energy services/resources at community level. As such, the project is an Eligible Project Type in line with the requirements.
2.1.3) Projects shall conform to the Principles and Requirements	The project conforms with the Principles and Requirements detailed in the document. The project is eligible under section 4, Principle 1, section (a) of the Principles and Requirements as it follows an established Gold Standard methodology. Concerning point 4.1.7, the project does not support geoengineering or entail energy production from fossil fuels or nuclear.
General Eligibility Criteria	
3.1.1 Types of Project Type of project: Section 3.1.1 (b) identifies "End-use energy efficiency" projects as eligible and uses an example of "efficient cooking".	This project activity meets this criterion as it is an efficient cookstove project activity. The project aims at distributing ICS to households. ICS have a higher thermal efficiency and then falls in the efficient cooking category.
3.1.2 Project Area, boundary and scale Project area, boundary and scale: Section 3.1.2 (b): Project Area and Boundary shall be defined in line with the applicable Impact Quantification Methodologies and Product Requirements.	The proposed project activity is a large scale energy efficiency project activity, and the project area as well as the boundaries are clearly defined in A.1. and section A.2 The project takes place inside geographical boundaries of the Republic of Mozambique. The project is estimated to achieve 216,283 tCO ₂ emission reduction and thus qualify as a large scale project.

3.1.4 Legal ownership a) Projects involving the distribution of a large number of devices for services shall provide a clear description of the ownership of the Products that are generated under Gold Standard Certification all along the investment chain. In line with FPIC requirement, the proofs that end-users are aware of and willing to give up their rights on Products shall be provided. b) The transfer for Product ownership shall be discussed during the local stakeholder consultations for projects.	a) The ownership is defined in section A.1.2 below. The stove recipients are clearly explained by the local partner about the transfer of the ownership rights and selling the emission reductions resulting from the project activity once they accept the stove. They are explained that by signing the Carbon Transfer Form and providing their contact information for monitoring purposes, they enter into agreement that the VERs are transferred to the project developer. b) The transfer of Product ownership was discussed during the local stakeholder consultation.
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No.	Eligibility criteria of GHG Emissions Reductions & Sequestration Product Requirements (Version 2.1).	Justification
1.	General Eligibility Criteria	This project claim credits for the Emission Reductions associated with the use of fuel-efficient cookstove distributed by the project which is an eligible component of the project. The project is not submitted as a bundled project.
2.	Location of Project	The project is located in Mozambique which is eligible project location as Gold Standard VER Projects may be located in any host country or state.
3.	Eligible Greenhouse Gases	Only the eligible greenhouse gases (CO ₂ , CH ₄ and N ₂ O) are accounted for GS crediting as indicated in the Section B.1 of this PDD. Mozambique do not have mandatory operational schemes to reduce GHG emissions in any form (e.g. cap & trade, carbon tax etc.).
4.	Types of Project	This project is a "End-Use Energy Efficiency Improvement" project and thus eligible to issue GS VERs in line with the para 2.1.9 of the GHG Emissions Reductions & Sequestration Product Requirements (Version 2.1).
5.	ODA Declaration	No ODA assistance is provided to the project under the condition that the credits generated by the Project will be transferred, either directly or indirectly, to the donor country providing ODA support.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

The project owner (Carbonsink) has full and uncontested legal ownership of the Verified Emission Reductions (VERs) that are generated under Gold Standard Certification which is demonstrated by the transfer of credit ownership which is formalized with full, prior and informed consent from cookstove end-users and cookstove producers. Each project stove end-users sign a written assertion giving their permission for the credit transfer at the time they buy the project stove. The written assertions of the same are also given by each stove producer supplying the project.

A.2 Location of project

Host Country: The Republic of Mozambique.

Province: Maputo Province

Specific area: Maputo and other urban areas within the Maputo Province like the city of Matola and districts of districts of Boane, Matutuine, Moamba, Marracuene, Namaacha, Manhica.

(Figure 1).

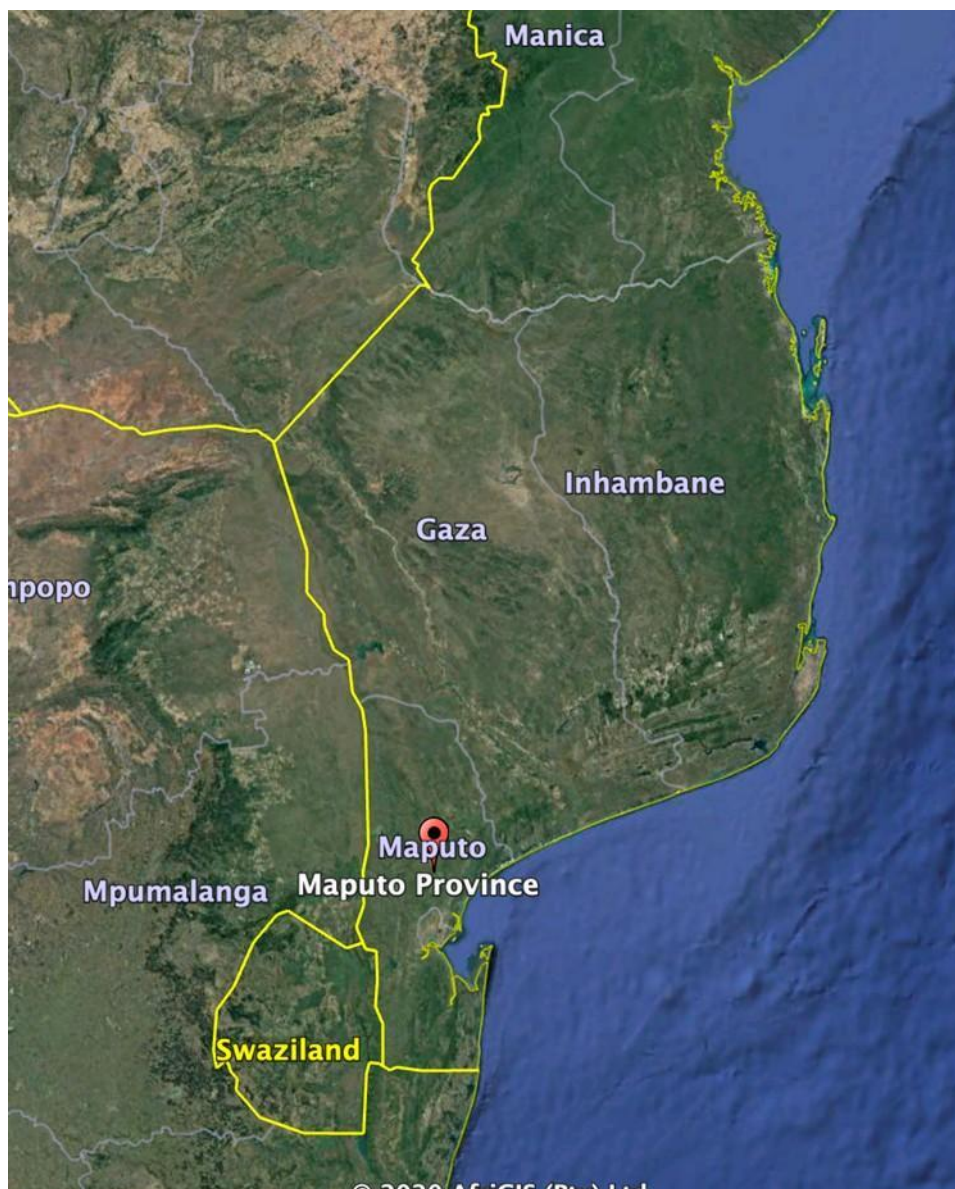


Figure 1 - Location of Maputo Province and City of Maputo within Mozambique.

A.3 Technologies and/or measures

The distributed stove model is locally manufactured Mbaula Poupa+ cookstove and it's composed of ceramic internal part and metallic (aluminium) outer skirt and base. The stoves will be manufactured by experienced local producers and using local and easily accessible materials³. This stove model is also known as Mbaula A o as Mbaula Alu. No structured warranty program is in place within the project activity, although the project

³ The useful lifetime of the stove model is estimated ex-ante as 5 years. For the ex-post ER estimates, the usage survey will establish a useful lifetime for technologies after which they are removed from the project database and no longer credited in line with the applied TPDDTEC ver 3.1 (page 31).

developer reserves the possibility to substitute damaged cookstoves with new devices of the same model once their usage is compromised.



Figure 2 - Mbaula Poupa+ cookstove

Table A -2 - The technical characteristics of the Mbaula Poupa+ stove model are resumed in the below table.

Mbaula Poupa+	
Average Thermal Efficiency	30.5 %
Fuel	Charcoal
Stove technology	Combustion
Single pot / multi pot	Single pot
Portable / fixed model	Portable
Unit Size (height x width x depth)	22.8 x 28.0 x 28.0 cm

Contribution to sustainable development

Besides reducing GHG emission in line with the UN's Sustainable Development Goal (SDG) number 13 this project activity will contribute also to the following SDGs:

- SDG 1: No poverty

The project activity includes the distribution of fuel-efficient cookstove technologies to replace the inefficient baseline cookstoves and thus result in reduction in charcoal consumption and related daily household expenditures for fuel. The provision of the technology at a subsidized price, coupled with the reduced fuel demand for the daily cooking activities, increases access to affordable energy and improved cooking technologies (i.e. basic services) for the stove users. Moreover, charcoal expenses are one of the major daily expenses of the local families and, therefore, a reduction in charcoal consumption, will reduce the end-user expenses related to the purchase of charcoal for cooking purposes.

- SDG 7: Ensure access to affordable energy

The distribution of fuel-efficient cookstove technologies to replace the inefficient baseline cookstoves as well as the provision of the fuel-efficient cookstove technology at a subsidized price, coupled with the reduced fuel demand for the daily cooking activities, increases access to affordable energy for the local community.

- SDG 15: Life on land

Mozambique has been suffering from massive loss of forest cover from 2001 to 2019, accounting for 3.29 MHa of tree cover, equivalent to a 11% decrease in tree cover since 2000, and 816Mt of CO₂ emissions⁴. Charcoal production is one of the major factors of deforestation and forest degradation and, therefore, it can be inferred that a reduction in charcoal demand created by this project (through the use of fuel-efficient project stoves), will turn in reduce deforestation/forest degradation and have a thus positive impact on soil conditions and biodiversity⁵.

A.4 Scale of the project

The project is a Large Scale project activity.

⁴ Global Forest Watch (site visited on 02/07/2020).

⁵ Sedano et al., 2016 - The impact of charcoal production on forest degradation: a case study in Tete. Sedano et al., 2020 - Monitoring forest degradation from charcoal production with historical Landsat imagery. A case study in southern Mozambique.

A.5 Funding sources of project

The project is funded by private sources of the Project Developer and with the financial contribution of AICS (Italian Agency for Development Cooperation). There are no official development assistance (ODA) funding for this project activity in exchange of the carbon credits. Please see the signed ODA declaration uploaded to SustainCERT platform.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

Gold Standard Methodology - Technologies and Practices to Displace Decentralized Thermal Energy Consumption Version 3.1, August 2017⁶

Requirements and Guidelines: Usage Rate Monitoring Version 2.0, 27 October 2020

Activity Requirements: [Community Services Activity Requirements v1.2](#)

B.2. Applicability of methodology (ies)

The choice of the selected methodology is justified by demonstrating (see the below table) that the project meets each applicability condition of the applied methodology.

Table B-1 - Applicability of the TPDDTEC methodology

Methodology applicability requirement	Justification regarding the project activity	Confirmation
This methodology is applicable to programs or activities introducing technologies and/or practices that reduce or displace greenhouse gas (GHG) emissions from the thermal energy consumption of households and non-domestic premises. The project technologies/practices must include introduction of improved biomass or fossil fuel cookstoves, ovens, dryers, space and water heaters (solar and otherwise), heat retention cookers or solar cookers.	This project activity includes introduction of new, fuel efficient, charcoal burning cookstoves to reduce the use of non-renewable biomass to meet thermal energy requirements for household cooking.	✓

⁶ Documents listed before the introduction of the new version 4.0 can still apply version 3.1. Please refer to annex GS11209 Approval of deviation request.

A project proponent implements the activity or Programme of Activities (PoA). The individual households and institutions do not act as project proponent.	The project activity is implemented by Carbonsink Group S.r.l. The individual households and institutions will not act as project participants.	✓
The project boundary needs to be clearly identified, and the technologies counted in the project are not included in another voluntary market or CDM project activity (i.e. no double counting takes place). In some cases, there maybe another similar activity within the same target area. Project proponents must therefore have a survey mechanism in place together with appropriate mitigation measures so as to prevent any possibility of double counting.	The project activity is implemented in Mozambique. The detailed project location is described the Section A.4.4 of this PDD. Each project stove has a unique identification number which will be recorded in the project database as outlined in the Section B.7.3 of this PDD. Moreover, the relevant online carbon registries have been checked to confirm that there are no risks for double counting.	✓
Technologies each have a continuous useful energy output of less than 150kW per unit (defines as total energy delivered usefully from start to end of operation of a unit divided by time of operation). For technologies or practices that do not deliver thermal energy in the project scenario but only displace thermal energy supplied in the baseline scenario, the 150kW	The continuous useful energy output of unit (i.e. cookstove) is less than 150kW as demonstrated in the Emission Reduction spreadsheet.	✓

threshold applies to the displaced baseline technology.		
The use of baseline technology as backup or auxiliary technology in parallel with the improved technology introduced by the project activity is permitted as long as a mechanism is put into place to encourage the removal of the old technology (e.g. discounted price for the improved technology) and the definitive discontinuity of its use. The project documentation must provide a clear description of the approach chosen and the monitoring plan must allow for a good understanding of the extent to which the baseline technology is still in use after the introduction of the improved technology. For example, whether the existing baseline technology is not surrendered at the time of the introduction of the improved technology, or whether a new baseline technology is acquired and put to use by targeted end users during the project crediting period. The success of the mechanism put into place must therefore be monitored and the approach must be adjusted if proven unsuccessful.	The mechanism to encourage end-users to abandon the use of the baseline technology is the selling of efficient stoves with a subsidized price and the informing of local communities about the benefits of the efficient stoves during the selling campaigns. Monitoring activities will study the extent of the baseline technology remaining in use and thereafter, the success of the mechanism put into the place, as described in the Section B.7.3 of this PDD. The ratio of end-user that are still cooking with baseline technology in the project scenario will be monitored and accounted for to calculate the fuel consumption and the corresponding project emissions as described in section B.6.	✓

If an old technology remains in use in parallel with the improved technology, the corresponding emissions must be accounted for as part of the project emissions.		
The project proponent must clearly communicate to all project participants the entity that is claiming ownership rights of and selling the emission reductions resulting from the project activity. For technology producers and the retailers of the improved technology or the renewable fuel in the project situation, this must be communicated by contract or clear written assertions in the transaction paperwork. If the claimants are not the project technology end-users, the end-users should be informed and notified that they cannot claim for emission reductions from the project. For example, forms signed by end-users alerting the end-users regarding the transfer of their carbon rights in exchange for pricing or the improved technology, which discounts its true cost.	The project representative and the project developer of this large scale project activity are the same entity (Carbonsink) and thus no need to separate legally-binding contractual agreement. The ownership and selling rights of the emission reductions resulting from the project activity are both held by Carbonsink. Each stove producer as well as stove end-buyer confirm the Carbon Transfer by signing an assertion. Moreover, the stakeholders were also informed about the transfer of carbon credit ownership during the stakeholder consultation process.	✓
Project activities making use of a new biomass feedstock in the project situation (e.g. shift from	No new biomass feedstock will be used in the project situation and	N/A

non-renewable to green charcoal, plant oil or renewable biomass briquettes) must comply with relevant Gold standard specific requirements for biomass related project activities, as defined in the latest version of the Gold standard rules ⁷ . If the biomass feedstock is sourced from a dedicated plantation, the criteria must apply to both plantations established for the project activity AND existing plantations that were established in the context of other activities but will supply biomass feedstock.	therefore this condition is not applicable.	
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Table B-2 - Applicability of Community Services Activity Requirements (version 1.2)

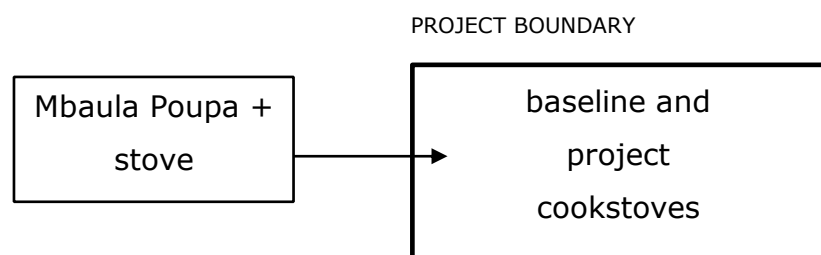
Eligibility criteria	Justification
Eligible project types	
Paragraph 2.1.2: All CSA Projects shall lead to climate change mitigation and/or adaptation by providing or improving access to services/resources at the household or community or institution level. Eligible services include electricity and energy, water and sanitation, waste management, housing, etc.	This project claims credits for the Emission Reductions associated with the use of fuel-efficient cookstoves distributed to local households. the pr

⁷ Gold Standard Toolkit, Annex C, and rule updates released prior to the time of first submission of the project activity to the Gold Standard.

Paragraph 2.1.3: Conformity to Principles & Requirements (and associated documents).	The project conforms with the Principles and Requirements as per Section A.2 and A.3.
General eligibility criteria	
Paragraph 3.1.1: Types of project	The project is an "End-use energy efficiency" project which is one of the pre-identified CSA project types noted in para 3.1.1. of the "Community Services Activity Requirements (Version 1.2)." "End-use energy efficiency" projects are defined as project activities that reduce energy requirements as compared to baseline scenario without affecting the level and quality of services or products, where the end-user of the products and services are clearly identified and when the physical intervention is required at the user end.
Paragraph 3.1.2: Project area, boundary and scale	The project is a Large Scale project activity.
Paragraph 3.1.3: Suppressed Demand Scenario	Not relevant for the project. No suppressed demand scenario shall be applied to the baseline scenario.
Paragraph 3.1.4: Legal ownership	As proved with the "Carbon Transfer Confirmations" uploaded to SC App, a clear description of the ownership of the Products (i.e. carbon credits) that are generated under Gold Standard Certification all along the investment chain is provided and end-users are aware of and willing to give up their rights. The transfer of credit ownership was discussed also during local stakeholder consultation.

B.3. Project boundary

The project boundary is the physical, geographical sites of the project technologies, in other words, the physical location cookstoves used within Maputo Province as described in above Section A.2. The sources and gases included in the project boundary are described in the below table.



Source		GHGs	Included?	Justification/Explanation
Baseline scenario	Heat delivery	CO ₂	Yes	Important source of emissions as CO ₂ is emitted any time charcoal is burned.
		CH ₄	Yes	Important source of emissions released during partial or incomplete combustion of charcoal.
		N ₂ O	Yes	Important source of emissions released during partial or incomplete combustion of charcoal.
Project scenario	Heat delivery	CO ₂	Yes	Important source of emissions as CO ₂ is emitted any time charcoal is burned.
		CH ₄	Yes	Important source of emissions released during partial or incomplete combustion of charcoal.
		N ₂ O	Yes	Important source of emissions released during partial or incomplete combustion of charcoal.

B.4. Establishment and description of baseline scenario

According to the applied TPDDTEC methodology, the baseline scenario is defined by the typical baseline fuel consumption patterns. A project activity may have multiple baseline scenarios that are applicable in relation to the different project technologies in the project activity depending on local fuel and technology use patterns. Furthermore, additional baseline scenarios can be added to a project activity at any time during the crediting period through a design change as per Gold Standard rules or alternatively, adjustment factors can be applied to existing baseline scenario to account for less significant variability in fuel consumption or technology, without the need to create a new baseline scenario.

The following paragraphs describes the baseline studies of this project.

A. Baseline Non-Renewable Biomass (NRB) assessment

The method used to develop the fraction of non-renewable biomass (fNRB) for Mozambique is carried out in line with Annex 1 section A.1.3 of TPDDTEC v.3.1. The source of data is the Global Forest Resources Assessment Reports for the years 2015⁸ and 2020⁹ from FAO.

$$a. \quad fNRB = NRB / (NRB + DRB)$$

$$86\% = 100,152,498 / (100,152,498 + 16,262,285)$$

where:

fNRB = Fraction of non-renewable biomass in the country (%)

NRB = Quantity of non-renewable biomass in the country (t/yr)

DRB = Demonstrably Quantity of renewable biomass in the country (t/yr). The approach satisfies the following condition regarding DRB:

Condition 1 - The woody biomass is originating from land areas that are forests where:

(a) The land area remains a forest; and

(b) Sustainable management practices are undertaken on these land areas to ensure, in particular, that the level of carbon stocks on these land areas does not systematically decrease over time (carbon stocks may temporarily decrease due to harvesting); and

(c) Any national or regional forestry and nature conservation regulations are complied with.

$$b. \quad NRB = R - DRB$$

$$100,152,498 = 116,414,783 - 16,262,285$$

And

$$c. \quad R = MAI + \Delta F$$

$$116,414,783 = 85,759,936 + 30,654,847$$

⁸ <http://www.fao.org/forest-resources-assessment/past-assessments/fra-2015/country-reports/en/>

⁹ <http://www.fao.org/forest-resources-assessment/fra-2020/country-reports/en/>

And

$$d. MAI = F * GR^{10}$$

$$85,759,936 = 36,743,760 * 2.33$$

Where:

R = Total annual biomass removals in the country (t/yr)

MAI = Mean Annual Increment of biomass growth (t/yr)

ΔF = Annual change in living forest biomass (t/yr)

F = Extent of forest (ha)

GR = Annual Growth rate of biomass (t/ha-yr)

$$e. DRB = PA * GR$$

$$16,262,285 = 6,967,560 * 2.33$$

Where:

PA = Protected Area Extent of Forest (ha)

B. Baseline Survey on target population characteristics

The baseline survey of target population characteristics is made with in person interviews with a robust sample of targeted end-users without project technologies that are representative of end-users targeted in the project activity. The baseline survey aims to provide information on target population characteristics, baseline technology use, fuel consumption, leakage and sustainable development indicators.

In line with the applied methodology, the survey should be carried out following the below guidelines for minimum sample size and was tailored to gather the following data:

Sample size

- Group size > 1000: Minimum sample size 100

Data collected

- Address or location and telephone number (mobile or landline where possible)
- The number of people served by the baseline technology and typical usage patterns and tasks (e.g. commercial, institutional, domestic, etc.)
- Types of baseline technologies used and estimated frequency
- Types of fuels used and estimated quantities
- Seasonal variations in baseline technology and fuel use
- Sources of fuels (purchased or hand-collected, etc.) and prices paid or effort made (e.g. walking distances, persons collecting, opportunity costs)

¹⁰ Country-specific growth rate calculated as a weighted average based on FAO reporting on distribution of total forest area by ecological zone and IPCC above-ground biomass growth rates for different ecological zones.

- Renewability and non-renewability indicators as required by Annex 1 of the applied TPDDTEC methodology

The Baseline Survey report was conducted between 18th May and 14th July 2020 in the urban areas of the cities of Maputo, Matola and Boane, in Maputo Province. The survey was originally performed for VPA titled "GS5658 VPA 22: Production and sale of efficient cookstoves in the urban areas of Maputo Province" and was meant to be applied also for future project activity implemented in the same area. Results of the survey are provided in a separate report¹¹.

C. Baseline Fuel Consumption

As the baseline fuel is charcoal the baseline fuel consumption is to be estimated through conduction of Baseline Performance Field Test. The Baseline Field Test (i.e. Kitchen Performance Test) is to be designed and conducted following the indications of paragraph 4 ("Baseline Studies"), Section 7 ("Performance Field Tests and Calculation of Emission Reductions") and Annex 4 ("Kitchen Performance Test") of the applied TPDDTEC methodology.

The baseline KPT should include a random sample of end users without project technologies that are representative of end users targeted in the project activity (minimum sample size is 20) and the statistical analysis should fulfil the requirements of the applied TPDDTEC methodology.

The results of the Baseline Kitchen Performance Test (KPT) implemented within the project area following the above-described indications are summarized below. The full description of the made baseline KPT is provided within a separate document¹².

Baseline KPT was implemented in the project area between May and July 2020 with sample size of 24 households

Average daily charcoal consumption resulted as 2.10 kg of charcoal/household/day. This value meets the 95/10 confidence/precision level of the applied TPDDTEC methodology.

According to the above, quantity of fuel that is consumed in the baseline, $P_{b,y} = 2.10$ kg/household-day.

B.5. Demonstration of additionality

The project activity is deemed additional as per the applied Community Services Activity Requirements (v 1.2) which deems additional any project located in LDC and – as per the Positive list) " project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions

¹¹ "GS10777 Baseline Survey Rerport 2020" uploaded to SustainCERT

¹² "GS10777_Baseline KPT Report_2020" uploaded to SustainCERT

and where each unit results in ≤ 600 MWh of energy savings per year or ≤ 600 tonnes of emission reductions per year” which is the case of this project.

B.5.1 Prior Consideration

N/A

B.5.2 Ongoing Financial Need

N/A

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)
13 Climate Action (mandatory)	N/A	Climate Change Mitigation measured as reduced tCO ₂ e ¹³
SDG 1	Target 1.4: By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance.	Indicator 1.4.1: Proportion of population living in households with access to basic services
SDG 7	Target 7.1: By 2030, ensure universal access to affordable, reliable and modern energy services	Indicator 7.1.2: Proportion of population with primary reliance on clean fuels and technology
SDG 15	Target 15.2: By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally	Indicator 15.2.1: Progress towards sustainable forest management

¹³ In line with the GG4GG Claims Guideline: “Unlike other SDGs the targets and Indicators associated with SDG13 cross-reference the Paris Agreement. A specific target is therefore not provided, Gold Standard interprets this as mitigation, measured as tCO₂e for the purpose of this document.”

B.6.1 Explanation of methodological choices/approaches for estimating the SDG Impact

Methodological choices/approaches related to SDG 13

The contribution to SDG 13 will be measured through carbon credits issued following the Gold Standard TPDDTEC v3.1 methodology.

Baseline Emissions

$$BE_{b,y} = B_{b,y} * ((f_{NRB} * EF_{b,fuel, CO2}) + EF_{b,fuel, nonCO2}) * NCV_{b, fuel}$$

Where:

$BE_{b,y}$	Emissions for baseline scenario b during the year y in tCO ₂ e
$B_{b,y}$	Quantity of fuel consumed in baseline scenario b during year y, in tons, as per by default factors (cases with project performance field test only)
f_{NRB}	Fraction of biomass used during year y for the considered scenario that can be established as non-renewable biomass
$NCV_{b,fuel}$	Net calorific value of the fuel that is substituted or reduced (IPCC default for wood fuel, 0.015 TJ/ton)
$EF_{b,fuel,CO2}$	CO ₂ emission factor of the fuel that is substituted or reduced. 112 tCO ₂ /TJ for Wood/Wood Waste, or the IPCC default value of other relevant fuel
$EF_{b,fuel,nonCO2}$	Non-CO ₂ emission factor of the fuel that is substituted or reduced

$$B_{b,y} = N_{p,y} * P_{b,y}$$

Where:

$N_{p,y}$	Project technology-days in the project database for project scenario p through year y
$P_{b,y}$	Specific fuel consumption for an individual technology in baseline scenario b during year y converted to tons/day

$$PE_{p,y} = B_{p,y} * ((f_{NRB} * EF_{p,fuel, CO2}) + EF_{p,fuel, nonCO2}) * NCV_{p, fuel}$$

Where:

$PE_{p,y}$	Emissions for project scenario p during year y in tCO ₂ e
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$B_{p,y}$	Quantity of fuel consumed in project scenario p during year y, in tons
fNRB	Fraction of biomass used during year y that can be established as non-renewable biomass
$NCV_{p,fuel}$	Net calorific value of the project fuel (IPCC default for Wood/Wood Waste 0.015 TJ/ton). This is equal to the baseline fuel NCV in projects which use the same fuel.

Emission Reduction Calculation

$$ER_y = \sum BE_{b,y} - \sum PE_{p,y} - \sum LE_{p,y}$$

Where:

ER_y	Emission reduction for total project activity in year y (tCO _{2e} /yr)
$BE_{p,y}$	Baseline emissions for baseline scenario b in year y (tCO _{2e} /yr)
$PE_{b,y}$	Project emissions for project scenario p in year y (tCO _{2e} /yr)
$LE_{p,y}$	Leakage for project scenario p in year y (tCO _{2e} /yr)

Methodological choices/approaches related to SDG 1

The project activity includes the distribution of fuel-efficient cookstove technologies to replace the inefficient baseline cookstoves and thus result in reduction in charcoal consumption and related fuel provision-related household expenditures. The provision of the technology at a subsidized price, coupled with the reduced fuel demand for the daily cooking activities, increases access to affordable energy and improved cooking technologies (i.e. basic services) for the stove users. Moreover, charcoal expenses are one of the major daily expenses of the local families and, therefore, a reduction in charcoal consumption, will reduce the end-user expenses related to the purchase of charcoal for cooking purposes.

The contribution of the project to this target of SDG 1 will be measured by calculating the average households reduction in expenditure for charcoal provision. This will be calculated as follows:

Average household savings i.e., decrease in expenditure on basic service such cooking, lighting, drinking:
 $(P_{b,y} - P_{p,y}) * \text{Fuel expenditure}$

Methodological choices/approaches related to SDG 7

The project activity includes the distribution of fuel-efficient cookstove technologies to replace the inefficient baseline cookstoves. The provision of the technology at a

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subsidized price, coupled with the reduced fuel demand for the daily cooking activities, increases access to affordable energy for the stove users.

The contribution of the project to this target of SDG 7 will be measured by calculating the number of active and eligible fuel-efficient cookstoves disseminated by the project, equal to the amount of households with increased access to energy thanks to the project activity. This will be calculated as follows:

$$N_{p,y} * U_{p,y} - \text{Distributed units} - N_{p,y}$$

Methodological choices/approaches related to SDG 15

Mozambique has been suffering from massive loss of forest cover from 2001 to 2019, accounting for 3.29 Mha of tree cover, equivalent to a 11% decrease in tree cover since 2000, and 816Mt of CO₂ emissions. The project includes the distribution of fuel-efficient cookstove technologies to replace the inefficient baseline cookstoves and thus result in reduction in charcoal consumption and production, which is one of the major factors of deforestation and forest degradation and, therefore, it can be inferred that a reduction in charcoal demand created by this project, will turn in reduce deforestation/forest degradation¹⁴.

The reduction in charcoal consumption will be demonstrated through the project KPT, compared with the baseline KPT.

$$fNRB * (P_{b,y} - P_{p,y}) * N_{p,y} * U_{p,y}$$

B.6.2 Data and parameters fixed ex ante

SDG13

Data/parameter	EF _{b,CO2}
Unit	tCO ₂ /tJ
Description	CO ₂ emission factor arising from use of fuels in baseline scenario
Source of data	IPCC default value
Value(s) applied	355.04
Choice of data or Measurement methods and procedures	Deemed valid by TPDDTEC Methodology
Purpose of data	Estimation of CO ₂ e emission reductions

¹⁴ Sedano et al., 2016 - The impact of charcoal production on forest degradation: a case study in Tete. Sedano et al., 2020 - Monitoring forest degradation from charcoal production with historical Landsat imagery. A case study in southern Mozambique.

Additional comment	Calculated for charcoal based on following: 1) IPCC default value for wood, 112 tCO ₂ /TJ (table 2.5 for default emission factors for stationary combustion) and 2) IPCC default value for wood to charcoal production ration = 112*3.17 = 355.04
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Data/parameter	EF _{b,non-CO2}				
Unit	tCO ₂ /TJ				
Description	Non-CO ₂ emission factor arising from use of fuels in baseline scenario				
Source of data	IPCC default value				
Value(s) applied	<table border="1"> <tr> <td>AR4</td><td>27.554</td></tr> <tr> <td>AR5</td><td>29.988</td></tr> </table>	AR4	27.554	AR5	29.988
AR4	27.554				
AR5	29.988				
Choice of data or Measurement methods and procedures	Deemed valid by TPDDTEC Methodology				
Purpose of data	Estimation of CO ₂ e emission reductions				
Additional comment	In line with the TPDDTEC methodology this term can include a combination of emission factors from fuel production, transport and use in line with the emission sources included in the project. And CO₂-emission factor for charcoal may be estimated by researching a conservative wood to charcoal ratio (for example IPCC) and multiplying this value by the pertinent EF for wood. Thus, here calculated based on following: 1) IPCC default value for wood, for CH₄: 0,3 tCO₂/TJ and for N₂O: 0,004 tCO₂/TJ (table 2.5 for default emission factors for stationary combustion), 2) GWP of CH₄ and N₂O for AR4 and AR5 and 3) IPCC default value for wood to charcoal production ration e.g. for AR4, calculation would be as follows: $(300/1000*25*3.17)+(4/1000*298*3.17) = 27.554$				

Data/parameter	EF _{p,co2}
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of fuels in project scenario
Source of data	IPCC default value for wood
Value(s) applied	355.04
Choice of data or Measurement methods and procedures	Deemed valid by TPDDTEC Methodology

Purpose of data	Estimation of CO ₂ e emission reductions
Additional comment	Calculated based on following: 1) IPCC default value for wood, 112 tCO ₂ /TJ (table 2.5 for default emission factors for stationary combustion) and 2) IPCC default value for wood to charcoal production ration = 112*3.17 = 355.04

Data/parameter	EF _{p,non co2}				
Unit	tCO ₂ /TJ				
Description	Non-CO ₂ emission factor arising from use of fuels in project scenario				
Source of data	IPCC default value for wood				
Value(s) applied	<table border="1"> <tr> <td>AR4</td><td>27.554</td></tr> <tr> <td>AR5</td><td>29.988</td></tr> </table>	AR4	27.554	AR5	29.988
AR4	27.554				
AR5	29.988				
Choice of data or Measurement methods and procedures	Deemed valid by TPDDTEC Methodology				
Purpose of data	Estimation of CO ₂ e emission reductions				
Additional comment	In line with the TPDDTEC methodology this term can include a combination of emission factors from fuel production, transport and use in line with the emission sources included in the project. And CO₂-emission factor for charcoal may be estimated by researching a conservative wood to charcoal ratio (for example IPCC) and multiplying this value by the pertinent EF for wood. Thus, here calculated based on following: 1) IPCC default value for wood, for CH₄: 0,3 tCO₂/TJ and for N₂O: 0,004 tCO₂/TJ (table 2.5 for default emission factors for stationary combustion), 2) GWP of CH₄ and N₂O for AR4 and AR5 and 3) IPCC default value for wood to charcoal production ration e.g. calculation for AR4 would be as follows: $(300/1000*25*3.17)+(4/1000*298*3.17) = 27.554$				

Data/parameter	NCV _b
Unit	TJ/ton
Description	Net calorific value of the fuels used in the baseline
Source of data	IPCC default value
Value(s) applied	0.0295

Choice of data or Measurement methods and procedures	Deemed valid by TPDDTEC Methodology, IPCC default value for charcoal
Purpose of data	Estimation of CO ₂ e emission reductions
Additional comment	

Data/parameter	NCV _p
Unit	TJ/ton
Description	Net calorific value of the fuels used in the project
Source of data	IPCC default value
Value(s) applied	0.0295
Choice of data or Measurement methods and procedures	Deemed valid by TPDDTEC Methodology, IPCC default value for charcoal
Purpose of data	Estimation of CO ₂ e emission reductions
Additional comment	This has same values as NCV _{baseline} in projects which reduce use of the same fuel.

B.6.3 Ex ante estimation of SDG Impact

Ex ante calculations related to the outcomes of SDG 13

The transparent ex-ante calculations of the outcomes of SDG 13 (i.e. CO₂e reductions) will be provided in a separate Excel Spreadsheet at the validation stage. For data/parameters available before design certification values contained in section B.6.2 and for data/parameters not available before design certification the estimates included in section B.7.1 have been used.

Below, examples of calculations are presented for the month of December, 2021 and relative forecasted number of eligible stoves:

Baseline Emissions, December 2021

$$BE_{b,2021} = B_{b,2021} * ((fNRB * EF_{b,fuel, CO_2}) + EF_{b,fuel, nonCO_2}) * NCV_{b, fuel}$$

$$B_{b,y} = N_{p,2021} * P_{b,2021}$$

$$BE_{b,2021} = (17,378 \text{ stoves} * 31 \text{ day} * 000210 \text{ tons/day/stove}) ((0.86 * 355.04 \text{ tCO}_2/\text{TJ}) + 29.988 \text{ tCO}_2/\text{TJ}) * 0.0295 \text{ TJ/ton}$$

$$BE_{b,2021} = 11,191 \text{ tCO}_2\text{e}$$

Project Emission, December 2021

$$PE_{p,2021} = B_{p,2021} * ((fNRB * EF_{p,fuel, CO2}) + EF_{p,fuel, nonCO2}) * NCV_{p, fuel}$$

$$B_{p,2021} = N_{p,2021} * ((P_{p,2021} * U_{p,2021}) + (P_{b,2021} * (1 - U_{p,2021})))$$

$$PE_{p,2021} = [17,378 \text{ stove} * 31 \text{ day} * ((0.000925 \text{ tons/day/stove} * 0.85) + (0.00210 \text{ tons/day/stove} * (1-0.85)))] * ((0.86 * 355.04 \text{ tCO}_2/\text{TJ}) + 29.988 \text{ tCO}_2/\text{TJ}) * 0.0295 \text{ TJ/ton}$$

$$PE_{p,2021} = 5,524 \text{ tCO}_{2e}$$

Emission reductions, December 2021

$$ER_y = \sum BE_{b,y} - \sum PE_{p,y} - \sum LE_{p,y}$$

$$ER_{2021} = 11,191 \text{ tCO}_{2e} - 5,524 \text{ tCO}_{2e} - 0 \text{ tCO}_{2e}$$

$$ER_{2021} = 5,667 \text{ tCO}_{2e}$$

Ex ante calculations related to the outcomes of SDG 1, SDG 7, and SDG 15

The transparent ex-ante calculations of the outcomes of SDG 1, SDG 7, and SDG 15 are provided in a separate Excel Spreadsheet and performed as per the GS SDG Impact Tool. For data/parameters available before design certification values contained in section B.6.2 and for data/parameters not available before design certification the estimates contained included in section B.7.1 have been used.

Below, examples of calculations are presented:

SDG 1:

Net benefit, year 1 (Average household savings))

$$(P_{b,1} - P_{p,1}) * \text{Fuel expenditure} = (2.1 - 0.91) * 18 * 365 = 7,782 \text{ MZN saved per household per day in Year 1}$$

SDG 7:

Net benefit, year 1 of Project Activity (households using efficient cookstove):

$$N_{p,1} * U_{p,1} - (\text{Distributed units} - N_{p,1}) = 12,020 * 0.9 - (14,220 - 12,020) = 10,186$$

B.6.4 Summary of ex ante estimates of each SDG Impact

SDG 13 – Emission Reductions (tCO_{2e})

Year	Baseline estimate	Project estimate	Net benefit
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Year 1	53,076	26,307	26,769
Year 2	161,735	81,420	80,315
Year 3	295,564	152,508	143,056
Year 4	339,724	183,656	156,068
Year 5	362,216	206,239	155,977
Total	1,212,315	650,131	562,185
Total number of crediting years	5		
Annual average over the crediting period	242,463	130,026	112,437

SDG 1 - Average household savings i.e., decrease in expenditure on basic service such cooking, lighting, drinking (MZN/hh. y)

Year	Baseline estimate	Project estimate	Net benefit
Year 1	13,797	6,015	7,782
Year 2	13,797	6,076	7,721
Year 3	13,797	6,138	7,659
Year 4	13,797	6,200	7,597
Year 5	13,797	6,262	7,535
Total	68,985	30,691	38,294
Total number of crediting years	5		
Annual average over the crediting period	13,797	6,138	7,659

SDG 7 – Increased access to energy (households using efficient cookstoves)

Year	Baseline estimate	Project estimate	Net benefit
Year 1	633	10,818	10,186
Year 2	1,769	28,572	26,803
Year 3	2,189	33,277	31,088
Year 4	2,489	35,473	32,983
Year 5	2,514	33,440	30,926

Total	2,514	35,473	32,983
Total number of crediting years	5		
Annual average over the crediting period	1,919	28,316	26,397

SDG 15 – Total non-renewable wood fuel saved (tonnes/y)

Year	Baseline estimate	Project estimate	Net benefit
Year 1	7,131	3,109	4,022
Year 2	18,834	8,295	10,540
Year 3	21,936	9,758	12,178
Year 4	23,383	10,507	12,876
Year 5	22,043	10,005	12,038
Total	93,328	41,674	51,654
Total number of crediting years	5		
Annual average over the crediting period	18,666	8,335	10,331

B.7. Monitoring plan

B.7.1 Data and parameters to be monitored

SDG 13

Data/parameter	$f_{NRB,y}$
Unit	Fractional of non-renewable biomass
Description	Non-renewability status of wood fuel during year y
Source of data	Refer to section B.4
Value(s) applied	0.86
Measurement methods and procedures	Calculation of f_{NRB} is performed as per Annex 1 of the applied methodology (v 3.1)
Monitoring frequency	N/A
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of project scenario

Additional comment	The value of f_{NRB} is fixed ex-ante for entire crediting period even though the project activity may choose to update the f_{NRB} during the crediting period.
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Data / Parameter	$P_{b,y}$
Unit	tons/household-day
Description	Quantity of fuel that is consumed in baseline scenario b during year y
Source of data	Baseline FT (Baseline KPT made 2020 ¹⁵), baseline FT updates, and any applicable adjustment factors
Value(s) applied	Estimated as 0.00210
Measurement methods and procedures	Field survey
Monitoring frequency	N/A
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Estimation of CO ₂ e emission reductions
Additional comment	In line with the TPDDTEC methodology a single baseline fuel consumption parameter is weighted to be representative of baseline technologies being compared for project crediting.

Data / Parameter	$P_{p,y}$
Unit	tons/household-day
Description	Quantity of fuel that is consumed in project scenario p during year y
Source of data	Ex post: Total sales record, Project FT, project FT updates, and any applicable adjustment factors Ex ante: Estimation is based on the difference between baseline fuel consumption $B_{b,y}$ and fuel saving in project scenario, the latter is calculated based on equation n. 2 from GS Simplified methodology (v 3.0), for which baseline stove efficiency is referenced as 12.5 %¹⁶ and project stove efficiency is estimates based on equation n. 3 from GS Simplified Methodology (v1.1) as described in Ex Ante ER Calculations.

¹⁵ Baseline Survey Report and KPT Report will be uploaded to SC App at validation stage

¹⁶ 'The Manica Charcoal Stove Project', Chris Bradnum, Journal of Energy in Southern Africa, Vol 24 No 2, 2013

Value(s) applied		
	Year	P_{p,y}
	1	0.000916
	2	0.000925
	3	0.000934
	4	0.000944
	5	0.000953
Measurement methods and procedures	Field survey	
Monitoring frequency	Updated every two years, or more frequently	
QA/QC procedures	Transparent data analysis and reporting	
Purpose of data	Estimation of CO ₂ e emission reductions	
Additional comment	In line with the TPDDTEC methodology a single project fuel consumption parameter is weighted to be representative of the quantity of project technologies of each age being credited in a given project scenario.	
	As a cross check between estimated fuel savings and actual fuel saved by each device, expected values for each project year are presented in Ex Ante Calculation Spreadsheet (tab 'Project Emission') and calculated thanks to expected decrease in project stove efficiency and shall be taken as referral during future verification.	

Data / Parameter	U _{p,y}
Unit	Percentage
Description	Usage rate in project scenario p during year y
Source of data	Ex post: Annual usage survey
	Ex ante: ex-ante estimation

Value(s) applied	<table> <tr> <th>Year</th><th>$U_{p,y}$</th></tr> <tr> <td>1</td><td>90%</td></tr> <tr> <td>2</td><td>85%</td></tr> <tr> <td>3</td><td>80%</td></tr> <tr> <td>4</td><td>75%</td></tr> <tr> <td>5</td><td>70%</td></tr> </table>	Year	$U_{p,y}$	1	90%	2	85%	3	80%	4	75%	5	70%
Year	$U_{p,y}$												
1	90%												
2	85%												
3	80%												
4	75%												
5	70%												
Measurement methods and procedures	Field survey												
Monitoring frequency	Annual or more frequently, in all cases on time for any request for issuance												
QA/QC procedures	Transparent data analysis and reporting												
Purpose of data	Estimation of CO _{2e} emission reductions												
Additional comment	A single usage parameter is weighted to be representative of the quantity of project technologies of each age being credited in a given project scenario as indicated in section II of the methodology. The project foreseen to follow both the mandatory and good practice monitoring requirements as indicated in the Annex 10 ("Rule Update: Requirements and Guidelines for carrying out usage surveys for projects implementing improved cooking devices") of the applied TPDDTEC methodology and therefore a maximum of 90% usage rate may be claimed. Parameter is also used to calculate: SDG 1 and SDG 7												
Data / Parameter	$N_{p,y}$												
Unit	Number of project cookstoves credited (units)												
Description	Technologies in the project database for project scenario p through year y												
Source of data	Total sales and distribution record												

Value(s) applied	Ex ante estimation:	
	Project year	N_{P,y}
	1	12,020
	2	33,613
	3	41,596
	4	47,296
	5	47,771
Measurement methods and procedures	Continuous monitoring. The number of project cookstoves is recorded in the stoves selling database.	
Monitoring frequency	Continuous	
QA/QC procedures	Transparent data analysis and reporting	
Purpose of data	Estimation of CO _{2e} emission reductions	
Additional comment	The total sales record is divided based on project scenario to create the project database. Note, that conservatively only 95% of the sold stoves is estimated eligible based on the baseline survey results. This percentage will be confirmed during the monitoring activities, indeed, a specific question will be included in the monitoring survey questionnaire to confirm the used baseline fuel.	
	Parameter is also used to calculate: SDG 1 and SDG 7	

Data / Parameter	LE _{p,y}
Unit	tCO _{2e} /year
Description	Leakage in project scenario p during year y
Source of data	Baseline and monitoring surveys
Value(s) applied	0
Measurement methods and procedures	Field survey
Monitoring frequency	Every two years
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Estimation of CO _{2e} emission reductions
Additional comment	Aggregate leakage can be assessed for multiple project scenarios, if appropriate

Data / Parameter	CU _b
Unit	tons/household-day
Description	Charcoal use in baseline scenario
Source of data	Baseline FT (Baseline KPT made 2020 ¹⁷), baseline FT updates, and any applicable adjustment factors
Value(s) applied	0.00210
Choice of data or Measurement methods and procedures	Field survey
Purpose of data	Estimation of CO ₂ e emission reductions Monitoring of SDG 3, SDG 12 and SDG 15
Additional comment	A single baseline fuel consumption parameter is weighted to be representative of baseline technologies being compared for project crediting. For this project a fixed baseline is adopted.

B.7.2 Sampling plan

The sampling approach is to be used for following surveys as summaries below:

Project Survey (PS) of end-user characteristics

- The Project Survey has the same requirements as the Baseline Survey (described in Section B.4 of this PDD) regarding the representativeness and sample sizing i.e. at least 100 households need to be sampled when the target groups size is > 1000.¹⁸
- When applicable the Project Survey and the Usage Survey will be implemented together using the same sample.

Usage Survey

- The minimum total sample size for Usage Survey is 100, with at least 30 samples for project technologies of each age being credited. The majority of interviews in a

¹⁷

¹⁸ Please refer to TPDDTEC V3.1 pag. 12

usage survey must be conducted in person. Thus, if technologies of age 1-5 are credited, the usage survey must include 30 representative samples from each age for total of 150 samples¹⁹. The resulting usage parameter should be weighted based on the proportion of technologies in the total sales record of each age.

- In line with the applied TPDDTEC methodology, any sampling methods can be used, provided that the sample is selected randomly. If sampling approach other than simple random sampling is applied, 'Guidelines for sampling and survey for CDM project activities and program of activities'²⁰ must be followed.
- Both the mandatory and good practice monitoring requirements as indicated in the Annex 10 ("Rule Update: Requirements and Guidelines for carrying out usage surveys for projects implementing improved cooking devices") of the applied TPDDTEC methodology shall be followed.
- When applicable the Usage Survey and the Project Survey will be implemented together using the same sample.

Leakage Assessment

Leakage assessment regarding the potential leakage sources will be completed every other year together with the Usage Survey and therefore no separate sampling approach are used for the leakage assessment.

Baseline and Project Performance Field Tests (BFT and PFT) i.e. Kitchen Performance Tests (KPT)

- Kitchen performance field test is carried out both for baseline and project scenarios, either by testing a paired sample (baseline and project performance measured for same subjects) or by independent sampling (different subjects, and usually different sample sizes, for baseline and project scenarios). The approach taken to conduct the performance tests must in any case be such that:
 - o it is transparent and can easily be replicated,
 - o it is evidently conservative,

¹⁹ Please refer to TPDDTEC V3.1 pag. 12

²⁰ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20151023152925068/Meth_GC48_%28ver04.0%29.pdf (site visited 23/06/2020)

- the sample is randomly selected so as to not introduce a material bias,
- and the impact of daily and seasonal variations on the expected average
- fuel consumption savings is accounted for.
- Representativeness: Fuel consumption must be measured with a representative sample of end users under each defined baseline scenario (in the absence of the project technology) and project scenario. For the representativeness the indications of the para 7 of the applied TPDDTEC (in page 17 of the methodology) are to be followed. The survey and field test can be conducted concurrently with the same subjects (for example, Project KPT can be conducted with the same subjects who participated in Usage/Project Survey).
- Sampling approach: Any sampling methods can be used, provided that the sample is selected randomly. If sampling approach other than simple random sampling is applied, Guidelines for sampling and surveys for CDM project activities and programme of activities must be followed.
- Sample sizing and statistical analysis:
 - The baseline and project field test data must be analyzed in combination to estimate the average annual emission reductions or average fuel savings per unit. As the baseline fuel and project fuel are the same (e.g. deployment of improved cookstove for the reduction of non-renewable biomass use), the statistical analysis can be conducted with respect to fuel savings per unit.
 - There are two valid options for the statistical analysis. In all cases, sample sizes must be greater than 20:
 - a. 90/30 rule. When the sample sizes are large enough to satisfy the “90/30 rule,” i.e. the endpoints of the 90% confidence interval lie within +/- 30% of the estimated mean, overall emission reductions can be calculated on the basis of the estimated MEAN annual emission reduction per unit or MEAN fuel annual savings per unit.
 - b. 90% confidence rule. When the sample sizes are such that the “90/30 rule” is not complied with, the emission or fuel saving result is not the mean (or average) test result, but a lower value²⁰, i.e. the LOWER BOUND of the onesided 90% confidence interval.

Refer to "GS10777_Baseline KPT Report_2020" for description of outcomes for BKPT and related data collection.

B.7.3 Other elements of monitoring plan

In line with the applied TPDDTEC methodology, the monitoring activities will involve data collection during distribution/installation as well as post-implementation information regarding the project cookstove technologies and their usage.

Firstly, the project proponent must keep records of the information collected in an electronic database including:

- Total sales record: The project proponent must maintain an accurate and complete sales record (or dissemination record in case of non-commercial distribution). The sales records must be maintained continuously and backed up electronically.
- Project database: The project proponent must maintain an accurate and complete project database. The project database is derived from the total sales record (or dissemination record in case of non-commercial distribution) with project technologies differentiated by different project scenarios. The differentiation of the project database into sections is based on the results of the applicable monitoring studies for each project scenario, in order that ER calculations can be conducted appropriately section by section. Technologies aged beyond their useful lifetime, as established in the usage survey, are removed from the project database and no longer credited.

Given that Usage Rate shall account for any drop off of either functionality of devices and/or decrease in adoption rate of technologies, eventual substitution activities shall be simply tracked in the project selling database. Given that substituted devices will belong to different age-groups (with different Usage Rates applied), drop offs shall be meant to be accounted in Usage Rate estimate for each single vintage.

No specific repairment/substitution program is in place within the project activity. Substitution of non-functional devices shall be incentivized within the selling campaigns and new devices will be grouped in age-groups²¹.

In line with the above-mentioned responsibilities, Carbonsink – as owner and implementer of the project activity – is in charge of data collection, quality control/assurance, training of selected monitoring team and management of data retrieved from field through the usage of Carbonsink Webapp.

Secondly, after the project implementation start, the project proponent must conduct the following on-going studies for each project scenario. These studies are foreseen to be made following both the mandatory and good practice monitoring requirements as indicated in the Annex 10 ("Rule Update: Requirements and Guidelines for carrying out usage surveys for projects implementing improved cooking devices")²² including thus, in addition to the mandatory requirements also organization and reporting of the field team training and supervision, end-user training and follow up visits as well as on awareness campaign.

a) Project Survey (i.e. Monitoring Survey) – Completed annually, beginning 1 year after project crediting period start

The monitoring survey investigates changes over time in a project scenario, and in a baseline scenario in case renewal of crediting period, by surveying end users with project technologies on an annual basis. It provides critical information on year-to-year trends in end user characteristics such as technology use, fuel consumption and seasonal variations.

b) Usage Survey – Completed annually, or more frequently, and in all cases on time for any request of issuance

The usage survey provides a single usage parameter that is weighted based on drop off rates that are representative of the age distribution for project

²¹ Refer to Ex Ante ER Estimates spreadsheet for further insights on how different usage rates are applied to different vintages.

²² The "Good Practice Monitoring Requirement" regarding the recording of the field team training and supervision, end-user training and follow ups as well as regarding the awareness campaign are foreseen to be followed.

technologies in the total sales record. The resulting usage parameter should be weighted based on the proportion of technologies in the total sales record of each age²³. This weighted usage parameter must be applied when calculating the quantity of fuel consumed in project scenario p during year y ($B_{p,y}$). Unless proven otherwise, it should be assumed that any drop off in the use of the project technology is replaced by fuel consumption in the applicable baseline scenario. The usage survey will establish a useful lifetime for technologies after which they are removed from the project database and no longer credited.

Moreover, the usage survey will study the eventual continued usage of the baseline stoves contemporary with the project stoves to ensure that the emission from the potential continued use of the baseline technologies will be accounted in the emission reduction calculations.

c) Project FT Update - Completed every other year, or more frequently

The PFT update is an extension of the project PFT and provides a fuel consumption assessment representative of project technologies currently in use every two years. Hence the PPT update accounts for changes in the project scenario over time as project technologies age and new customers are added, also as new models and designs are introduced. Moreover, as PFT (i.e. Kitchen Performance Test) will measure the average total fuel usage per households it will automatically capture also the eventual fuel consumption caused by the contemporary usage of baseline cookstoves.

d) Baseline FT Update

For this project a fixed baseline is adopted.

e) Leakage Assessment - Completed every other year, starting on time for the first verification

²³ For example, if only technologies in the first year of use (age0-1) are being credited, a usage parameter must be established through a usage survey for technologies age0-1. If an equal number of technologies in the first year of use (age0-1) and second year of use (age1-2) are credited, a usage parameter is required that is weighted to be equally representative of drop off rates for technologies age0-1 and age1-2

Leakage assessment will be conducted together with Usage Survey. The following potential sources of leakage are to be assessed:

- The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or in a manner suggesting more usage than would have occurred in the absence of the project.
- Non-project users who previously used lower emitting energy sources use the non-renewable biomass or fossil fuels saved under the project activity.
- The project significantly impacts the NRB fraction within an area where other CDM or VER project activities account for NRB fraction in their baseline scenario.
- The project population compensates for loss of the space heating effect of inefficient technology by adopting some other form of heating or by retaining some use of inefficient technology.
- By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution within households who commonly used a technology with relatively lower emissions, in cases where such a trend is not eligible as an evolving baseline.

If the leakage assessment quantifies an increase in fuel consumption by the nonproject households/users attributable to the project activity, then calculations must be adjusted to account for the quantified leakage. Leakage is either calculated as a quantitative emissions volume (tCO₂e) or as a percentage of total emission reductions.

Non-Renewable Biomass Assessment Update

- In line with the applied TPDDTEC methodology the non-renewable biomass fraction may remain fixed for the entire crediting period. In case of a renewal of the crediting period and as per Gold Standard rules, the NRB fraction must be reassessed as any other baseline parameters and updated in line with most recent data available.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1 Start date of project

01/08/2020

The above mentioned date is the date when the first project stoves were sold.²⁴

Project is regular, in line with GS4GG Principle 4.

C.1.2 Expected operational lifetime of project

15 years, i.e. 180 months

C.2. Crediting period of project

C.2.1 Start date of crediting period

02/08/2020

C.2.2 Total length of crediting period

5 years. The crediting period may be renewed twice in line with Community Services Activity Requirements.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1 Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 1. Human Rights	N/A
Principle 2. Gender Equality	N/A
Principle 3. Community Health, Safety and Working Conditions	N/A

²⁴ Please refer to document "GS11209_Project start date" uploaded to SustainCERT

Principle 4.1 Sites of Cultural and Historical Heritage	N/A
Principle 4.2 Forced Eviction and Displacement	N/A
Principle 4.3 Land Tenure and Other Rights	N/A
Principle 4.4. Indigenous people	N/A
Principle 5. Corruption	N/A
Principle 6.1 Labour Rights	N/A
Principle 6.2 Negative Economic Consequences	N/A
Principle 7.1 Emissions	N/A
Principle 7.2 Energy Supply	N/A
Principle 8.1 Impact on Natural Water Patterns/Flows	N/A
Principle 8.2 Erosion and/or Water Body Instability	N/A
Principle 9.1 Landscape Modification and Soil	N/A
Principle 9.2 Vulnerability to Natural Disaster	N/A
Principle 9.3 Genetic Resources	N/A
Principle 9.4 Release of pollutants	N/A
Principle 9.5 Hazardous and Non-hazardous Waste	N/A
Principle 9.6 Pesticides & Fertilisers	N/A
Principle 9.7 Harvesting of Forests	N/A
Principle 9.8 Food	N/A
Principle 9.9 Animal husbandry	N/A
Principle 9.10 High Conservation Value Area sand Critical Habitats	N/A
Principle 9.11 Endangered Species	N/A

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

A.1.

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?

The project reflects the key issues and requirements of gender sensitive design and implementation: the project activities provide access to fuel efficient cookstoves for local communities. Especially women and girls, mainly responsible on fuel collection and cooking activities, will benefit of the use of fuel efficient cookstoves. The project will not reduce or put at risk women's access to or control of resources or benefits. Instead, it's foreseen that women, as main responsible form cooking fuel resources and cooking activities will have better control of these resources (quantity of wood needed/day is reduced and time will be saved as the cooking may be made faster) and moreover to benefit most for the expected health improvements caused by the reduced smoke inhalation during the cooking activities. Moreover, the already implemented activities related to the Stakeholder Consultation (for example, the cooking demonstration made during the stakeholder consultation meetings) as well as the future implementation of the project activities takes into the account gender roles and the abilities of women and men to participate in the decision/designs of the project activities. The more detailed description of the Stakeholder Consultation made in the project design phase and its design in a way which enabled both women and men participating in the consultation meeting are described in the separate Stakeholder Consultation Report.

Question 2 - Explain how the project aligns with existing country policies, strategies and best practices

Project activities are in line with the country policies, strategies, and best practices. In fact, the project activities are foreseen to improve the daily conditions of the local families, especially woman and girls by providing access to efficient cooking technology. The access to improved cooking technology, in

	addition to the contribute on reduction of smoke inhalation during the cooking activities, will reduce the need of fuel for daily cooking activities and consequently will reduce the effort for charcoal supplying.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	Not required. The safeguarding principles and requirements assessment of this project do not include particular challenges regarding gender perspective and, thus, it has been enough to address gender issue following the default requirements of the GS4GGs.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	Not required. The stakeholder consultation of this project do not include particular challenges regarding gender perspective and, thus, it has been enough to address gender issue following the default requirements of the GS4GGs.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1 Summary of stakeholder mitigation measures

No specific concerns were raised during the stakeholder consultation and thus no mitigation measures were identified. Please refer to annex GS11209_SC Report, Section C.4.

E.2 Final continuous input / grievance mechanism

	Method Chosen (include all known details e.g. location of book, phone number, identity of mediator)	Justification of Choice (best practice)
Continuous Input / Grievance Expression Process Book (mandatory)	Carbonsink's office at Rua do Telegrafo, 109 Maputo, Mozambique	This is the address of both project implementer's office in Maputo. Th participants agreed that it is appropriate to consider to place th book for eventual comments and grievance.
GS Contact (mandatory)	help@goldstandard.org	

Telephone access (optional)	+ 258 82 536 2620 (Carbonsink) + 258 84 016 0634 (Nova Energia)	This method was found to be useful also because it can help stakeholders to make their contributions in case of difficulties mobility. Many end-users, in general, own a phone/smartphone
Internet/email access (optional)	info@carbonsink.it help@goldstandard.org	These electronic addresses were shared during the meeting for stakeholders with access to emails
Nominated Independent Mediator (optional)	Not indicated	It was decided to not nominate an independent mediator
Other	Not indicated	N/A

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	The project is implemented respecting internationally proclaimed human rights and is no complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights. The project doesn't either discriminate with regards to participation and inclusion as the efficient project stoves are free and are distributed for the families selected in collaboration with the representatives of the local communities	N/A

Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	No	JUSTIFICATION POINT 1: The project activity doesn't endorse any form of discrimination based on gender. The cookstoves can be brought by anyone living in the project area. It's not foreseen that the Project reduces or put at risk women's access to or control of resources, entitlements and benefits. Instead, it's foreseen that the women, as main responsible for charcoal supply and cooking activities will have better control of resources (fuel and time will be saved) and moreover to benefit most for the possible health improvements caused by the reduced smoke inhalation during the cooking activities. It is thereafter not either foreseen that the Project would adversely affect man and	N/A

		women in marginalised or vulnerable communities. There will be less burden on women, men and children, as less charcoal for cooking needs to be purchased. This will reduce the time burden on women and men in the socially isolating activity of collecting resources. The Project takes into account gender roles and the abilities of women and men to participate in the decision/designs of the project activities. For example, the Stakeholder Consultation made in the project design phase included both women and men participating in the consultation meeting. In fact, women's participation and engagement in the project (as they are the main responsible of the cooking activities) will be essential for the success of the whole project.	
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		The Project will take into account gender roles and the abilities of women and men to participate and benefit from the project activities. For example, the training/cooking demonstrations on using the new stoves and on their benefits will be targeted especially for women who are traditionally responsible for the cooking activities. The project is not contributing to an increase in women's workload or preventing them from engaging in other activities. In fact, the efficient cookstoves will reduce the charcoal need for daily cooking activities and will thereafter reduce women's and girls' workload related to the cooking fuel supply. The project is not foreseen to reproduce or depend discrimination against women. The women's role will be	
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		essential as the cookstove users and they will have the possibility for giving feedback regarding the project as any other community member. The project is not foreseen to limit women's ability to use, develop and protect natural resources. Instead, the use of the efficient cookstoves will reduce the charcoal consumption and will thereafter give a possibility for saving and local natural wood resources. The project activity will not expose women or girls to further risks or hazards. Instead, the risk related to the smoke inhalation during the cooking activities or the risks related to the chrcoal supplying are foreseen to be reduced. JUSTIFICATION POINT 2: The Project will not directly or indirectly lead or contribute to	
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		adverse impacts on gender equality or the situation of women. In fact, the use of the efficient project cookstoves is foreseen to improve the general conditions of women and not to lead to any risk of contributing issues like sexual harassment/ exploitation, violence, human trafficking slavery, imprisonment, drudgery or restriction of women's rights or access to resources. The Project will not have any impact on women's ownership rights to inherit and own land, homes and other assets. The Project applies the principles of nondiscrimination and equal treatment and equal pay for equal work. For the project monitoring activities and for any other eventual paid or volunteer work the principle of the equal	
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		pay for equal work will be applied and it will be organized in way to provide the conditions for equitable participation of men and women whenever possible. Project activities like using the efficient cookstoves is not having any limitations on participation or benefiting from the Project depending on the pregnancy, maternity/paternity leave or marital status. JUSTIFICATION POINT 3: The Project will not include any specific gender related risks. The “National Action Plan on Women, Peace and Security” of Mozambique has been consulted to assess this risk. JUSTIFICATION POINT 4: Not applicable.	
Principle 3. Community Health, Safety and Working Conditions			

1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	No	The project activities doesn't include exposing the community to increased health risks and is not adversely affecting the health of workers and the community. In fact, the project activities provide the distribution of improved cookstoves, with the aim, omitting the other objectives, to improve the health of households, for example through the reduction of smoke and therefore less harmful inhalations.	N/A
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project activity doesn't include sites, structures or objects with historical, cultural, artistic, traditional or religious value or intangible forms of culture.	N/A
>>			
Principle 4.2 Forced Eviction and Displacement			

Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The Project will provide improved cookstoves to the households in the project area and it does not require alternation, damage or removal of any historical, artistic, traditional, religious or cultural heritage issues.	N/A
>>			
Principle 4.3 Land Tenure and Other Rights			
a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	a. The project activity consists of distributing improved cookstoves and therefore no physical or economic relocation of peoples is involved. The project activity consists of distributing improved cookstoves and therefore no physical or economic relocation of peoples is involved.	N/A
b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?			
>>		b. Stoves distribution doesn't need additional lands to be used and, therefore, doesn't require any change to land tenure arrangements and/or other rights. In fact, the aim of the project is to reduce the	

		quantity of charcoal consumed in the project area.	
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	There is no interference between the project activity and claiming of land tenure by indigenous people. Also, no indigenous people are present in the Maputo city area.	N/A
>>			
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	The Project doesn't involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects. The project is, in fact, implemented in the field by in collaboration with Carbonsink. The ethical codes of both	N/A

		entities are against corruption. Moreover, Mozambique has ratified the UN Convention against Corruption.	
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours	No	The project is implemented in the field by Carbonsink. The employees' rights are a cross-cutting issue and respected in all of the projects of project partners. Mozambique has ratified ILO Convention 87 and 98. All employees, mainly hired for production and distribution of the cookstoves (in the first phase) and for monitoring activities, will work voluntarily for the project, no forced labour is used and all employment is in compliance with national laws and consistency with the principles of standard ILO conventions. Indeed, Mozambique has ratified ILO Convention 29 and 105.	N/A

per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion)		The workers can establish and join labour organizations. In case of new workers will be hired, the working agreement will be documented and implemented in compliance with the Section 3.6.1 of GS4GG Safeguarding Principles & Requirements version 1.2. The employment model applied will be also locally and culturally appropriate. Moreover, all the possible staff hired has a minimum age of 18. Mozambique has ratified ILO Convention 138 and 182. The use of the efficient cookstove will reduce the quantity of charcoal used in daily cooking activities and can thereafter release economical and time resources of the local families for other tasks which can be considered to support the financial sustainability of the project.	
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5.	The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures		The use of efficient cookstoves will reduce the charcoal consumption and will thereafter save the resources of the project families and can be considered to have positive impacts on the project families' economic situation.	
Principle 6.2 Negative Economic Consequences				
1.	Does the project cause negative economic consequences during and after project implementation?	No	No potential risks for the local economy are expected.	N/A
>>				
Principle 7.1 Emissions				
	Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	The Project will reduce the GHG emissions as will be monitored and verified in line with the GS4GG.	N/A
>>				
Principle 7.2 Energy Supply				

Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The Project will not use energy from a local grid or power supply. The efficient cookstoves are fired with charcoal and therefore no change for the currently used cooking fuel will be made.	N/A
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project will not affect the natural or pre-existing pattern of watercourses, groundwater and/or the watersheds etc. water related issues.	N/A
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	The Project will not cause additional erosion directly or indirectly and/or water body instability or disrupt the natural pattern of erosion.	N/A

b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability?		The project area is not susceptible to excessive erosion and/or water body instability.	
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The project impact on environment is positive, no negative impacts are expected. Moreover, the stove distribution activities does not include planting or other agricultural activities, producing chemicals or use of GMOs. The project will distribute one clay stove model produced locally. The local stove production is not having any significant environmental impacts as for example the quantities of clay collection needed for the stove production are low compared to other activities like house construction. Neither hazardous waste is produced.	N/A
>>			

Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	Furthermore, the aim of the project is to reduce the quantity of charcoal consumed in the project area for cooking activities which will save the natural resources. The Project is not suspected to or will lead to increased vulnerability to any extreme climatic conditions.	N/A
>>			
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	The Project doesn't involve or is negatively impacted by the use of genetically modified organisms or GMOs.	N/A
>>			
Principle 9.4 Release of pollutants			

Could the Project potentially result in the release of pollutants to the environment?	No	The Project is not potentially resulting in release of pollutants to the environment.	N/A
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	The Project is not involving the manufacture, trade, release, and/or use of hazardous chemicals and or materials.	N/A
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	The Project doesn't involve the application of pesticides and/or fertilisers.	N/A
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	Neither harvesting of forests is involved.	N/A
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality	No		

of food available such as through crop regime alteration or export or economic incentives?		The Project doesn't modify the quantity or nutritional quality of food available.	
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The Project doesn't involve animal husbandry.	N/A
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project boundary is the physical, geographical sites of the project technologies, in other words, the physical location of the project stoves.	N/A
>>		1.	2.
Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)?	No	There are no endangered species identified as potentially being present the project boundary and the project is not foreseen to have any negative potential impacts on other	N/A

AND/OR		areas where endangered species may be present through transboundary affects.	
Does the Project potentially impact other areas where endangered species may be present through transboundary affects?			
>>			

Appendix 2- Contact information of project participants

Organization name	Carbonsink Group S.r.l. (Carbonsink)
Registration number with relevant authority	
Street/P.O. Box	Piazza Beverini
Building	4
City	La Spezia
State/Region	
Postcode	19121
Country	Italy
Telephone	+39 055 4574675
E-mail	info@carbonsink.it
Website	www.carbonsink.it
Contact person	Gianluca Persia
Title	
Salutation	Mr.
Last name	Persia
Middle name	
First name	Gianluca
Department	
Mobile	
Direct tel.	+39 055 4574675
Personal e-mail	gianluca.persia@carbonsink.it

APPENDIX 3- LUF ADDITIONAL INFORMATION

N/A

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to Design Change [Requirements](#) for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption